House Price Prediction — Project Summary

Project Overview

The objective of this project was to build a machine learning model that can predict house prices based on different property features such as area, number of bedrooms, bathrooms, location-related features, and available facilities. The dataset used for this project was the Housing Prices Dataset, which contains information about residential properties.

Data Processing

The dataset was loaded using Pandas and explored to understand its structure, features, and target variable. Data cleaning was performed by checking for missing values, removing duplicate records, and converting categorical variables into numerical values using one-hot encoding. The cleaned dataset was then prepared for machine learning by separating the features and the target variable (price).

Model Development and Evaluation

Two regression models were trained and compared: Linear Regression and Random Forest Regressor. The dataset was divided into 80% training data and 20% testing data. The Linear Regression model achieved an MAE of approximately 970,043, RMSE of approximately 1,324,507, and an R^2 score of 0.653. The Random Forest model achieved an MAE of approximately 1,021,546, RMSE of approximately 1,400,566, and an R^2 score of 0.612. Based on these results, Linear Regression performed better for this dataset.

Key Findings

The analysis showed that factors such as house area, number of rooms, bathrooms, air conditioning, and preferred location features have a strong impact on house prices. The correlation analysis helped identify relationships between different features and the final price. The model was able to predict house prices with moderate accuracy and explained around 65% of the variation in prices.

Business Recommendation

Real estate businesses should focus on accurately evaluating important property features such as size, location, and available facilities while setting prices. Highlighting premium features like air conditioning and preferred locations can help attract buyers and improve pricing strategies. Using machine learning models can also support faster and more data-driven property valuation decisions.